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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It sorts data using a single pass
c. It is less efficient than bubble sort
d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Heap sort
b. Insertion sort
c. Bubble sort
d. Quick sort

The correct answer is:
Insertion sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys

b. A sorting algorithm is stable if it preserves the order of all keys

c. A sorting algorithm is stable if it preserves the order of duplicate keys

d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **4**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists

b. It uses the heap data structure

c. It bubbles up the largest element in each pass

d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates

b. It makes the analysis slower

c. It complicates the analysis process

d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It complicates the selection of items
- c. It slows down the process
- d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are ignored
- c. The parts are merged without sorting
- d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Quick sort
- c. Heap sort
- d. Selection sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Sorting data in a single pass
c. Rearranging data without sorting
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are combined to form a single sorted list
c. They are split again into smaller sublists
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()
b. order()
c. arrange()
d. sort()

The correct answer is: sorted()

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. order()
b. arrange()
c. sorted()
d. sort()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Quick Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Merge Sort
c. Quick Sort
d. Linear Search

The correct answer is: Merge Sort